100% INTERVIEW READY

THE 20 TIME SERIES PATTERNS YOU NEED TO MASTER

Don't memorize queries. Master the exact date/time patterns used by **Netflix**, **Uber** & **Amazon**.

STEAL THESE PATTERNS →

Swipe to start →

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Problem 1/20

DATE DIFF

AMAZON

Slow Shipping > 7 Days

Identify Service Level Agreement (SLA) breaches where delivery took more than a week.

SAMPLE INPUT

order_id	ordered	delivered
101	2026-01-01	2026-01-06
102	2026-01-01	2026-01-10

EXPECTED OUTPUT

order_id	days_taken
102	9

SOLUTION

MySQL

```mysql
SELECT order_id,
  DATEDIFF(delivered, ordered) as days_taken
FROM orders
WHERE DATEDIFF(delivered, ordered) > 7;
```

Next: Day Filter →

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THE 20-PATTERN SQL CHEAT SHEET

SAVE

REPOST

Problem 2/20

DAY FILTER

DOMINO'S

Orders on Fridays

Filter all order records that occurred specifically on a Friday using DAYNAME.

SAMPLE INPUT

order_id	order_date
101	2026-02-13
102	2026-02-14
103	2026-02-20

EXPECTED OUTPUT

order_id	order_date
101	2026-02-13
103	2026-02-20

SOLUTION

MySQL

```mysql
SELECT *
FROM orders
WHERE DAYNAME(order_date) = 'Friday';
```

Next: Format Anchor →

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THE 20-PATTERN SQL CHEAT SHEET

SAVE

REPOST

Problem 3/20

FORMAT ANCHOR

REPORTING

First Day of Month

Generate a reporting anchor date by forcing the day part to '01'.

SAMPLE INPUT

curr_date
2026-02-16

EXPECTED OUTPUT

month_start
2026-02-01

SOLUTION

MySQL

```mysql
SELECT
  DATE_FORMAT(curr_date, '%Y-%m-01') as month_start
FROM report_dates;
```

Next: Day of Year →

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THE 20-PATTERN SQL CHEAT SHEET

SAVE

REPOST

Problem 4/20

DAY OF YEAR

CALENDAR

Day Number in Year

Track annual progress by converting a date to its day number (1-365).

SAMPLE INPUT

order_id	order_date
500	2026-01-01
501	2026-02-01

EXPECTED OUTPUT

order_id	day_num
500	1
501	32

SOLUTION

MySQL

```mysql
SELECT
  order_id,
  DAYOFYEAR(order_date) as day_num
FROM orders;
```

Next: Is Null →

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SAVE

REPOST

Problem 5/20 IS NULL

LOGISTICS

Pending Orders

Identify backlog orders where the delivery date has not been set yet.

SAMPLE INPUT

id	delivery_date
1	2026-02-10
2	NULL

EXPECTED OUTPUT

id	delivery_date
2	NULL

SOLUTION

MySQL / SQL Server

```sql
SELECT *
FROM orders
WHERE delivery_date IS NULL;
```

Next: Date Format →

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THE 20-PATTERN SQL CHEAT SHEET

SAVE REPOST

Problem 6/20

DATE FORMAT

EUROPE OPS

Localization (dd/mm)

Format dates for European reporting standards (Day/Month/Year).

SAMPLE INPUT

order_id	order_date
10	2026-02-16

EXPECTED OUTPUT

order_id	eu_date
10	16/02/2026

SOLUTION

MySQL

```mysql
SELECT
  order_id,
  DATE_FORMAT(order_date, '%d/%m/%Y') as eu_date
FROM orders;
```

Next: Global Agg →

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THE 20-PATTERN SQL CHEAT SHEET

SAVE

REPOST

Problem 7/20

GLOBAL AGG

ANALYTICS

Avg Daily Orders

Calculate the global throughput: Total Orders divided by Total Distinct Days.

SAMPLE INPUT

total_orders	distinct_days
100	5

EXPECTED OUTPUT

avg_daily_orders
20

SOLUTION

MySQL / SQL Server

```sql
SELECT
  COUNT(order_id) / COUNT(DISTINCT DATE(order_date))
  as avg_daily_orders
FROM orders;
```

Next: Intersect →

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THE 20-PATTERN SQL CHEAT SHEET

SAVE

REPOST

Problem 8/20 INTERSECT

SPOTIFY

Active 2022 AND 2023

Find users who placed orders in BOTH years (Year-over-Year retention).

SAMPLE INPUT

user_id	order_date
A	2022-05-10
A	2023-08-15
B	2022-11-20

EXPECTED OUTPUT

user_id
A

SOLUTION

MySQL

```mysql
SELECT DISTINCT user_id FROM orders WHERE YEAR(order_date) = 2022
AND user_id IN (
  SELECT DISTINCT user_id FROM orders WHERE YEAR(order_date) =
2023
);
```

Next: Date Calc →

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THE 20-PATTERN SQL CHEAT SHEET

SAVE REPOST

Problem 9/20

DATE CALC

OUTLOOK

Most Recent Monday

Calculate the date of the most recent Monday relative to the current date.

SAMPLE INPUT

curr_date
2026-02-18

EXPECTED OUTPUT

last_monday
2026-02-16

SOLUTION

MySQL

```mysql
SELECT SUBDATE(curr_date, WEEKDAY(curr_date))
 as last_monday
FROM calendar;
-- Assumes Monday is index 0
```

Next: Floor Math →

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THE 20-PATTERN SQL CHEAT SHEET

SAVE

REPOST

Problem 10/20 FLOOR MATH

HISTORY

Orders Per Decade

Group data by decades (1990, 2000, 2010) using integer math on the Year.

SAMPLE INPUT

order_date
1995-06-15
2001-08-20
2009-12-31

EXPECTED OUTPUT

decade_start	cnt
1990	1
2000	2

SOLUTION

MySQL

```mysql
SELECT
  FLOOR(YEAR(order_date) / 10) * 10 as decade_start,
  COUNT(*) as cnt
FROM orders
GROUP BY decade_start;
```

Next: Date Math →

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THE 20-PATTERN SQL CHEAT SHEET

SAVE REPOST

Problem 11/20

DATE MATH

FEDEX

Overnight Delivery

Count orders where the delivery happened exactly one day after the order date.

SAMPLE INPUT

order_date	delivery_date
2026-02-01	2026-02-02
2026-02-01	2026-02-03

EXPECTED OUTPUT

overnight_count
1

SOLUTION

MySQL

```mysql
SELECT COUNT(*) as overnight_count
FROM orders
WHERE DATEDIFF(delivery_date, order_date) = 1;
```

Next: Min Aggregation →

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THE 20-PATTERN SQL CHEAT SHEET

SAVE

REPOST

Problem 12/20

MIN AGG

SAAS METRICS

Time to First Purchase

Calculate the average onboarding velocity (days between signup and first order).

SAMPLE INPUT

u_id	signup	1st_order
1	2026-01-01	2026-01-05
2	2026-01-01	2026-01-03

EXPECTED OUTPUT

avg_velocity
3.0

SOLUTION

MySQL

```mysql
SELECT AVG(DATEDIFF(first_order, signup_date)) as avg_velocity
FROM (
  SELECT u.user_id, u.signup_date, MIN(o.order_date) as
first_order
  FROM users u JOIN orders o ON u.user_id = o.user_id
  GROUP BY u.user_id
) sub;
```

Next: Row Number →

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THE 20-PATTERN SQL CHEAT SHEET

SAVE

REPOST

Problem 13/20 | ROW NUMBER | TEMU

Last 3 Orders Per User

Retrieve the 3 most recent orders for every single customer using Window Functions.

SAMPLE INPUT

u_id	order_id
1	104
1	103
1	102
1	101

EXPECTED OUTPUT

u_id	order_id	rn
1	104	1
1	103	2
1	102	3

SOLUTION

MySQL / SQL Server

```sql
SELECT * FROM (
  SELECT *, ROW_NUMBER() OVER(PARTITION BY user_id ORDER BY
  order_date DESC) as rn
  FROM orders
) sub
WHERE rn ≤ 3;
```

Next: DAU Metric →

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THE 20-PATTERN SQL CHEAT SHEET

SAVE | REPOST

Problem 14/20

DAU METRIC

FACEBOOK

Daily Active Users

Calculate DAU by counting distinct user IDs for every single date.

SAMPLE INPUT

order_date	user_id
2026-01-01	A
2026-01-01	A
2026-01-01	B

EXPECTED OUTPUT

activity_date	dau
2026-01-01	2

SOLUTION

MySQL / SQL Server

```sql
SELECT
  DATE(order_date) as activity_date,
  COUNT(DISTINCT user_id) as dau
FROM orders
GROUP BY activity_date;
```

Next: Null Check →

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THE 20-PATTERN SQL CHEAT SHEET

SAVE

REPOST

Problem 15/20

NULL CHECK

LINKEDIN

Dormant Since Signup

Users who signed up more than 30 days ago but have NEVER placed an order.

SAMPLE INPUT

user_id	signup	order_id
A	2020-01-01	NULL
B	2026-02-14	NULL

EXPECTED OUTPUT

user_id
A

SOLUTION

MySQL

```
SELECT u.user_id
FROM users u
LEFT JOIN orders o ON u.user_id = o.user_id
WHERE o.order_id IS NULL
AND u.signup_date < CURRENT_DATE - INTERVAL 30 DAY;
```

Next: Timestamp Diff →

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THE 20-PATTERN SQL CHEAT SHEET

SAVE

REPOST

Problem 16/20

TIMESTAMP DIFF

UBER

Exact Time (Minutes)

Measure delivery performance with high precision by calculating minutes elapsed.

SAMPLE INPUT

order_date	delivery_date
2026-02-16 10:00:00	2026-02-16 10:45:00

EXPECTED OUTPUT

mins_to_deliver
45

SOLUTION

MySQL

```mysql
SELECT
  TIMESTAMPDIFF(MINUTE, order_date, delivery_date)
  as mins_to_deliver
FROM orders
WHERE delivery_date IS NOT NULL;
```

Next: ISO Yearweek →

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THE 20-PATTERN SQL CHEAT SHEET

SAVE

REPOST

Problem 17/20

ISO YEARWEEK

AIRBNB

Orders by Year-Week

Standard weekly reporting using the YEARWEEK function to group high-volume data.

SAMPLE INPUT

order_date
2026-01-05
2026-01-06
2026-01-12

EXPECTED OUTPUT

yyyyww	total_orders
202601	2
202602	1

SOLUTION

MySQL

```mysql
SELECT
  YEARWEEK(order_date) as yyyyww,
  COUNT(*) as total_orders
FROM orders
GROUP BY yyyyww;
```

Next: Moving Average →

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THE 20-PATTERN SQL CHEAT SHEET

SAVE

REPOST

Problem 18/20 | MOVING AVG | ROBINHOOD

3-Month Moving Avg

Smooth out revenue trends by calculating the average of the current month plus previous 2.

SAMPLE INPUT

month	revenue
2026-01	10000
2026-02	20000
2026-03	30000

EXPECTED OUTPUT

month	moving_avg
2026-03	20000

SOLUTION

MySQL

```sql
SELECT month,
  AVG(revenue) OVER (
    ORDER BY month
    ROWS BETWEEN 2 PRECEDING AND CURRENT ROW
  ) as moving_avg
FROM monthly_sales;
```

Next: Group By Month →

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THE 20-PATTERN SQL CHEAT SHEET

SAVE REPOST

Problem 19/20

GROUP BY MONTH

NETFLIX

Monthly Order Freq

Profile user behavior by counting how many orders each user places per month.

SAMPLE INPUT

user_id	order_date
A	2026-01-01
A	2026-01-15
B	2026-01-05

EXPECTED OUTPUT

user_id	month	cnt
A	2026-01	2
B	2026-01	1

SOLUTION

MySQL

```sql
SELECT
  user_id,
  DATE_FORMAT(order_date, '%Y-%m') as month,
  COUNT(*) as cnt
FROM orders
GROUP BY user_id, month;
```

Next: Same-Day Join →

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THE 20-PATTERN SQL CHEAT SHEET

SAVE

REPOST

Problem 20/20

SAME-DAY JOIN

INSTACART

Converted on Signup

Count unique users who placed their first order on the exact same date they signed up.

SAMPLE INPUT

u_id	signup	order_date
A	2026-01-01	2026-01-01
B	2026-01-01	2026-01-02

EXPECTED OUTPUT

converted_count
1

SOLUTION

MySQL

```mysql
SELECT COUNT(DISTINCT u.user_id) as converted_count
FROM users u
JOIN orders o ON u.user_id = o.user_id
WHERE DATE(u.signup_date) = DATE(o.order_date);
```

Next: MISSION ACCOMPLISHED →

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THE 20-PATTERN SQL CHEAT SHEET

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MISSION ACCOMPLISHED

You're Interview Ready!

20 PATTERNS MASTERED 100%

- ✓ DATE DIFF
- ✓ DAY FILTER
- ✓ FORMAT ANCHOR
- ✓ DAY OF YEAR
- ✓ IS NULL
- ✓ DATE FORMAT
- ✓ GLOBAL AGG
- ✓ INTERSECT
- ✓ DATE CALC
- ✓ FLOOR MATH
- ✓ DATE MATH
- ✓ MIN AGG
- ✓ ROW NUMBER
- ✓ DAU METRIC
- ✓ NULL CHECK
- ✓ TIMESTAMP DIFF
- ✓ ISO YEARWEEK
- ✓ MOVING AVG
- ✓ GROUP BY MONTH
- ✓ SAME-DAY JOIN
- ♥ Follow for more

Found this helpful?

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